

Community Intelligence: Multi-Signal Contributor Credibility Scoring and Semantic Consensus Synthesis for Technical Question Answering

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Abstract—The ability to make informed technical decisions, share experiences, and exchange insights has made online Community Question Answering (CQA) platforms essential for modern developers. However, expert responses often remain unnoticed or buried under inaccurate, misleading, or outdated answers. Motivated by this limitation, we present Community Intelligence (CI), a framework that provides reliable technical answers integrated with multi-signal trust indicators and community consensus. By evaluating contributor credibility and clustering semantic agreement, CI synthesizes evidence-backed solutions with confidence metrics and visual decision flowcharts. Empirical evaluation demonstrates that CI significantly improves answer reliability and user trust compared to standard search and zero-shot LLMs.

Index Terms—Community Question Answering (CQA), Information Reliability, Trust Indicators, Semantic Clustering, Software Engineering Platforms.

I. INTRODUCTION

Community Question Answering (CQA) platforms—including Stack Overflow, GitHub Discussions, and technical forums—serve as essential cloud-based knowledge repositories where modern developers host, manage, and share technical solutions. Within these threaded discussion environments, technical accuracy and content quality depend heavily on peer review, user reputation, and crowd-sourced moderation [1][2][4].

As reliance on crowdsourced technical knowledge has grown, concerns regarding data accuracy, misinformation, and voting manipulation have intensified [9][10]. To safeguard information quality without imposing prohibitive manual moderation burdens, traditional platforms have historically relied on basic vote-based ranking, badge systems, and simple heuristic filtering [5][7]. However, empirical evaluations reveal that basic voting and heuristic filters often obscure high-quality expert

insights, leaving valuable solutions buried beneath inaccurate, outdated, or misleading responses [8][11].

To overcome these limitations, recent research explores hybrid approaches that combine crowdsourced credibility signals with neural models and Large Language Models (LLMs) [6][12]. While zero-shot generative LLMs offer rapid answers, they frequently introduce severe code hallucinations, unverified dependencies, and incorrect technical assumptions [21]–[23][25]. Consequently, contemporary architectures integrate semantic consensus clustering [13][14] with adaptive Retrieval-Augmented Generation (RAG) and iterative grounding techniques [17]–[20].

By coupling developer reputation metrics [1][3] with LLM-driven semantic agreement and hallucination verification [24][26], hybrid Community Intelligence (CI) frameworks strike a balance between automated synthesis and community-verified accuracy. This multi-signal approach drastically reduces computational overhead while delivering reliable, evidence-backed technical solutions [15][16][27].

II. LITERATURE SURVEY

Community Question Answering (CQA) platforms serve as vital cloud-based repositories where developers host, manage, and share software engineering solutions. Within these environments, technical accuracy depends heavily on user reputation, peer review, and crowdsourced moderation. However, concerns regarding data accuracy, voting manipulation, and outdated information have intensified as online developer communities expand. To mitigate manual moderation overhead, platforms historically relied on heuristic ranking, upvoting, and badge distributions. Empirical evaluations reveal that basic vote-based ranking often obscures high-quality expert

TABLE I
COMPARATIVE STUDY OF KEY RELATED WORKS

Ref #	Authors (Year)	Techniques Used	Pros	Cons
[1]	Sun <i>et al.</i> (2025)	GitHub interaction network graph mining + NLP on commit text	Outperforms basic follower metrics; derives fine-grained expertise scores	Restricted to single-platform (GitHub); static non-adaptive snapshot
[2]	Hamer <i>et al.</i> (2025)	Multi-metric benchmarking (commit history, account age, org affiliation)	Significantly improves dependency-trust classification accuracy	Specific to package ecosystems; unlinked to CQA answer content quality
[3]	Zolduoarrati <i>et al.</i> (2025)	Large-scale regression analysis on demographic/geographic SO cues	Quantifies systematic regional biases in community reputation points	Observational/correlational only; provides no bias-correction algorithm
[8]	Kabir <i>et al.</i> (2024)	Manual evaluation + linguistic analysis of ChatGPT vs. SO	Highlights that 52% of LLM answers contain factual/code errors	Purely observational; offers no automated hallucination detection model
[9]	Da Silva <i>et al.</i> (2025)	Empirical reliability analysis + community posting volume tracking	Quantifies post-ChatGPT platform activity drop (~20–26%)	Offers no mechanism to preserve expert retention or aggregate answers
[12]	Semantic Study (2025)	LLM text embeddings + Gaussian Mixture Model (GMM) clustering	Substantially outperforms TF-IDF/word2vec for sparse technical text	Dependent on single-embedding runs; lacks multi-run consensus checks
[14]	Topic Study (2023)	Recursive consensus clustering + word-embedding enrichment	Eliminates topic instability and oscillation common in single runs	Evaluated on generic social media/tweets, not technical code contexts
[17]	RAG Study (2025)	Benchmarking BM25, dense, and hybrid retrieval for code generation	Demonstrates BM25 + simple similarity fusion yields best stability	Operates on static code repositories rather than evolving CQA streams
[20]	Jeong <i>et al.</i> (2025)	Adaptive RAG triggered by internal model uncertainty signals	Reduces unnecessary retrieval calls while cutting LLM hallucinations	Uncertainty signal is self-contained, lacking crowd/community verification
[21]	Manakul <i>et al.</i> (2023)	Sampling-based self-consistency checking across LLM outputs	Detects hallucinations in black-box models without external KBs	Grounded solely on model self-agreement, not verified human solutions
[24]	Consensus Study (2026)	Static code sampling-consistency with MRCM similarity metric	Detects hallucinated code accurately without requiring code execution	Evaluates consensus across LLM samples only, ignoring crowd ground truth
[26]	De-Hallucinator (2024)	Iterative retrieval-grounding loop querying external API docs	Successfully corrects API misuse hallucinations via iterative re-prompts	Grounded on static documentation rather than live crowd-voted answers

insights, leaving critical solutions buried beneath inaccurate or misleading answers.

Researchers have formulated advanced metrics to derive developer technical expertise and contributor reputation across open-source ecosystems using logarithmic profiling, GitHub network mining, and demographic analysis [1][2][3][4]. Further systematic mapping and predictive analytics studies have identified structural patterns in content quality and answer rejection dynamics within crowdsourced platforms [5][6][7].

In recent years, researchers have explored combining crowdsourced credibility signals with neural architectures and Large Language Models (LLMs). Empirical evaluations comparing ChatGPT to Stack Overflow demonstrate that crowd answers remain vital due to frequent LLM correctness issues and leaderboard voting manipulation vulnerabilities [8][9][10][11]. While zero-shot generative LLMs deliver immediate responses, they frequently introduce code hallucinations, unverified package dependencies, and incorrect syntactic assumptions [21][22][23][25].

To resolve these limitations, contemporary frameworks leverage text clustering techniques, such as recursive consensus clustering and TF-IDF semantic grouping, to enable interpretable topic detection [12][13][14]. Furthermore, advances in multi-hop Retrieval-Augmented Generation (RAG), knowledge graph construction, and structured planning reduce hallucinations by grounding generative responses in retrieved external context [15][16][17][18][19][20]. Recent specialized models incorporate black-box sampling, package hallucination detection, and iterative grounding to quantify and mitigate code generation errors [24][26][27]. By coupling multi-signal log-normalized developer credibility metrics with semantic agreement clustering and automated visual synthesis, hybrid Community Intelligence (CI) architectures strike an effective balance between automatic knowledge extraction and community-verified technical reliability. The synthesized com-

parison of representative foundational works is summarized in Table I.

III. PROPOSED SYSTEM

The proposed *Community Intelligence* (CI) framework employs a hybrid computational architecture that bridges multi-signal contributor profiling, semantic consensus clustering, and AI-driven synthesis. Traditional CQA mechanisms rely primarily on raw upvote counts, which frequently obscure authoritative expert solutions beneath early or aesthetically upvoted posts [8][11]. Conversely, standalone zero-shot Large Language Models (LLMs) often generate plausibly written yet inaccurate code, introducing synthetic hallucinations and unverified dependencies [21][25]. The CI framework addresses these challenges by grounding AI-driven knowledge synthesis within multi-dimensional crowd credibility signals and mathematical semantic consensus.

The technical architecture of the Community Intelligence framework comprises four primary operational phases:

A. Multi-Signal Contributor Credibility Profiling

To move beyond basic vote counts, the system computes a log-normalized credibility score $S_{\text{cred}}(a, u)$ for each candidate answer a authored by user u . The engine evaluates six distinct platform signals: answer vote count, user account reputation, accepted answer flag, total silver/gold badges, and total historical answer contributions. Each raw metric is log-transformed to mitigate extreme distribution outliers and

weighted according to its empirical reliability signal:

$$\begin{aligned}
S_{\text{cred}}(a, u) = & 0.35 \cdot \text{Norm}(\log(1 + \text{Votes})) \\
& + 0.25 \cdot \text{Norm}(\log(1 + \text{Reputation})) \\
& + 0.15 \cdot \text{AcceptedFlag} \\
& + 0.10 \cdot \text{Norm}(\log(1 + \text{GoldBadges})) \\
& + 0.10 \cdot \text{Norm}(\log(1 + \text{TotalAnswers})) \\
& + 0.05 \cdot \text{Norm}(\log(1 + \text{SilverBadges}))
\end{aligned} \tag{1}$$

Based on the synthesized score, authors are categorized into four transparent credibility tiers: *Expert* ($\geq 65\%$), *Trusted* ($\geq 45\%$), *Regular* ($\geq 25\%$), and *Novice* ($< 25\%$).

B. Semantic Consensus Clustering and Support Weighting

To detect peer agreement and identify consensus across distinct answer threads, candidate text bodies $\{T_1, T_2, \dots, T_k\}$ are vectorized using Term Frequency-Inverse Document Frequency (TF-IDF) over unigram and bigram feature spaces:

$$\mathbf{v}_i = \text{TF-IDF}(T_i) \tag{2}$$

Pairwise semantic proximity between answer vectors is calculated via Cosine Similarity:

$$\text{Sim}(T_i, T_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|} \tag{3}$$

Answers satisfying a threshold condition ($\text{Sim}(T_i, T_j) \geq \theta = 0.40$) are grouped into semantic clusters C_k . A cluster is classified as a *Consensus Cluster* if supported by at least two distinct user contributors ($|\{u \mid (a, u) \in C_k\}| \geq 2$). The collective weight $W(C_k)$ of each cluster is scaled by combining average contributor credibility with cluster size:

$$W(C_k) = \left(\frac{1}{|C_k|} \sum_{(a,u) \in C_k} S_{\text{cred}}(a, u) \right) \times |C_k| \tag{4}$$

This formulation ensures solutions backed by multiple high-reputation developers receive priority over isolated responses.

C. Generative Synthesis and Visual Analytics

Once top-ranked consensus clusters are identified, the framework routes aggregated text along with contributor trust metrics into a Generative AI synthesis pipeline. Rather than generating ungrounded responses, the model operates strictly as an evidence-backed synthesizer. It transforms raw developer threads into structured schemas comprising:

- **Headline Consensus Solution:** A concise, definitive answer backed by overall confidence indicators.
- **Decision Flowcharts:** Visual execution flows rendering conditional pathways for developer implementation.
- **Comparative Solution Matrices:** Side-by-side trade-off evaluations (e.g., performance vs. complexity).
- **Code Blueprints & Anti-Pattern Alerts:** Production-ready code blocks paired with explicit warnings against deprecated or insecure community practices.

D. Asynchronous Architecture and Deployment

The framework is deployed as an asynchronous microservice powered by FastAPI and Uvicorn. Client queries initiate asynchronous data retrieval from the external Stack Exchange API. Candidate answers, user profiles, and voting histories are processed concurrently through the credibility and clustering engines, reducing pipeline latencies. Interactive web interfaces render real-time confidence scores, interactive flowcharts, and empirical benchmark comparisons against baseline search models.

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